

EXPERIMENT 04

Analyze Email Headers and Detect Email Spoofing Using MHA (Mail Header Analyzer)

OBJECTIVE

To access, extract, and analyze raw email internet headers, trace routing MTA servers via **Received:** fields, verify email authentication mechanisms (SPF, DKIM, DMARC), and evaluate potential email spoofing.

TOOLS / REQUIREMENTS

- **Email Platform / Client:** Gmail Web Interface
- **Analysis Tools:** Text Editor (Notepad) / Visual Email Header Inspection
- **Online Verification Utilities:** IP Geolocation & WHOIS Lookup (WhatIsMyIP / LookIP.net)

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

An institutional email communication was analyzed to determine authenticity, trace transmission infrastructure, and identify potential spoofing indicators. The investigator accessed the RFC 822 / MIME header, parsed MTA intermediate routing hops, executed IP lookup on origin servers, and verified cryptographic SPF, DKIM, and DMARC alignment.

EVIDENCE / INPUT

- **Target Evidence File:** GATE 2027 & JAM 2027 Examinations - Reg.eml
- **Sender Address:** "Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
- **Recipient Address:** csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
- **Message-ID:** <CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=c0bjpLXg@mail.gmail.com>
- **Delivery Time:** Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
- **Sending MTA Host IP:** 209.85.220.69 / 209.85.220.41

PROCEDURE

Step 1 — Accessing the Email in Mail Client

The target email ("Fwd: GATE 2027 & JAM 2027 Examinations - Reg") was opened in the Gmail interface to inspect sender headers, delivery details, and message context.

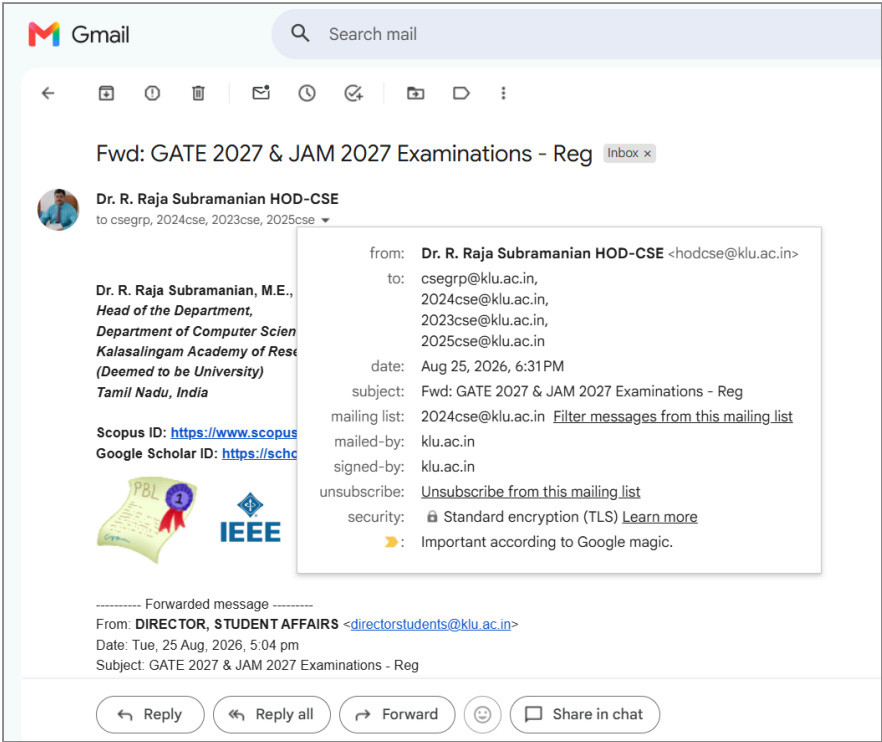


Figure 4.1: Target email accessed in mail client

Step 2 — Viewing Original Message Metadata

The *Show original* option was selected from the message menu, opening the RFC header overview displaying Message ID, creation timestamp, From, and To fields.

Original Message	
Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObj
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025

Figure 4.2: Original message metadata overview

Step 3 — Inspecting Mail Authentication Overview

The authentication overview table was evaluated:

Authentication Check	Evaluation Status	Details
SPF	PASS	Permitted sender IP 209.85.220.69
DKIM	PASS	Domain klu.ac.in signature verified
DMARC	PASS	Policy compliant

Original Message

Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObjpLXg@mail.gmail.com>
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: GATE 2027 & JAM 2027 Examinations - Reg
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more
DMARC:	'PASS' Learn more

Figure 4.3: Email authentication status table (SPF, DKIM, DMARC)

Step 4 — Extracting Full RFC 822 Raw Headers

The complete original raw MIME header was displayed with *Download Original* and *Copy to clipboard* controls for detailed offline forensic parsing.

Original Message

Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObjpLXg@mail.gmail.com>
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: GATE 2027 & JAM 2027 Examinations - Reg
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more
DMARC:	'PASS' Learn more

[Download Original](#)
[Copy to clipboard](#)

```

Delivered-To: 99240041270@klu.ac.in
Received: by 2002:a05:7300:e113:b0:315:b785:cfb with SMTP id hd19csp4359716dyb; Tue, 25 Aug 2026 06:01:36 -0700 (PDT)
X-Forwarded-Encrypted: i=5; AHgh+RrscZupKotijL4NS719ImE2lRPGAB5LonFlzddL8Pz04Uv0i1qdyKCUvjHkiDh8bIVgdsAU2AcZCaUxg/w=@klu.ac.in
X-Received: by 2002:a05:6402:505b:b0:698:af31:5a9b with SMTP id 4fb4d7f45d1cf-6a42f1e08c0mr38071921a12.9.1787662895622; Tue, 25 Aug 2026 06:01:35 -0700 (PDT)
ARC-Seal: i=4; a=rsa-sha256; t=1787662895; cv=pass; d=google.com; s=arc-20260327; h=MA5FAA7xG61Fo0GEVp2Ho1i3MdlvYKPM8Xc1NW4-Tu02THFdh3i58akKwuv5P71u0K

```

Figure 4.4: Full original RFC 822 MIME headers interface

Step 5 — Analyzing Primary Receiving MTA Hop

The raw header was pasted into a text editor (email-header.txt). The top Received: line (Received: by 2002:a05:7300:e113:b0:315:b785:cfb with SMTP id...) was examined to confirm the final delivering MTA and ARC security seal.

```

Delivered-To: 99240041270@klu.ac.in
Received: by 2002:a05:7300:e113:b0:315:b785:cfb with SMTP id hd19csp4359716dyb; Tue, 25 Aug 2026 06:01:36 -0700 (PDT)
X-Forwarded-Encrypted: i=5; AHgh+RrsCZupKotI3L4NS719ImE2lRPGAB51OnFlzddL8Pz04U0V1qdyKCUVjHkIdH8bIVgdsAU2ACzCaUxg/w==@klu.ac.in
X-Received: by 2002:a05:6402:505b:b0:698:af31:5a9b with SMTP id 4fb4d7f45d1cf-6a42f1e08c0mr38071921a12.9.1787662895622; Tue, 25 Aug 2026 06:01:35 -0700 (PDT)
ARC-Seal: i=4; a=rsa-sha256; t=1787662895; cv=pass; d=google.com; s=arc-20260327; b=W45FaAZG61eOQGFyP2Hoi13WdUYKPMgRXgINV+Ju02IHfDbjSig8aKkuvw5P71uqK 2oLzUtEjg1MlFmZ5ANGTO+AEHz014trulvUjuelTk8pxQT/EayVohP61AB0VcRzG5G 5wbnHIUIqjMrUcLvrRHg67XhmyI0DYL5mv4ASurxYFVT6ZTQCFD40J3/eqKpkMIHwAsB rcdXxYpSRNG/Fr9154KQ7ff3vIyJhuteK/41Yp+2tTgcddZVE0FCTT20R/dK2HEWZj jhkO/dTO2E/y2 +dAVBdXwCzYoA1H1W30/6IKs9bClpv+iq9MDAwKqy69HY214PquKsZ hdaO==
ARC-Message-Signature: i=4; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:to:subject:message-id:date:from:in-reply-to :references:mime-version:dkim-signature; bh=xamsobc8g+24HJQFSf09vaVhNkH49/GRIZricuWAAM=; fh=Th9v4pSU48wv4z2HhpN33SBOZTP8dnv7WqCNKfUJ5K0=; b=001EK5Mn2Fm48/s3ca192S0z7kgZwdjdxYETXf6A3b1ljQp5DT1T4xh7kZmfVHKbz 1yBf1Wd5E5QAMuXjTBNeyzPob/VsZa9/uSrMoVtu7jUPKq+2P3Zj3jBlx8FQ43CgBBeV ol/zM2xqSol7l6m9H78LlrczJWd1Fy57ElECaksgCehAbojboxYfFzJzFugpHQziqFU rJsc3z0716RN52xhxvz0k6CRKyZ4ZfScf1x1315TNRq0AWykoWN/IEMTz2P1l+Im yohck6jTfAFNHyHFHrPVYwyqyFexJP37BSu7WqTGD107Ztg6RRHfiYuStj+4ub5r76 e1rw==; dara=google.com
ARC-Authentication-Results: i=4; mx.google.com; dkim-pass header.i=@klu.ac.in header.s=google header.b=Jo0p3hi8; arc-pass (i=3 spf=pass spfdomain=klu.ac.in dkim=pass dkdomain=klu.ac.in
  
```

Figure 4.5: Examination of top Received header hop

Step 6 — Reviewing Comprehensive Header Stream

The entire structured header stream was analyzed to verify the chronological relay sequence across intermediate mail transfer agents.

```

Received: by 2002:a05:7300:e113:b0:315:b785:cfb with SMTP id hd19csp4359716dyb; Tue, 25 Aug 2026 06:01:36 -0700 (PDT)
X-Forwarded-Encrypted: i=5; AHgh+RrsCZupKotI3L4NS719ImE2lRPGAB51OnFlzddL8Pz04U0V1qdyKCUVjHkIdH8bIVgdsAU2ACzCaUxg/w==@klu.ac.in
X-Received: by 2002:a05:6402:505b:b0:698:af31:5a9b with SMTP id 4fb4d7f45d1cf-6a42f1e08c0mr38071921a12.9.1787662895622; Tue, 25 Aug 2026 06:01:35 -0700 (PDT)
ARC-Seal: i=4; a=rsa-sha256; t=1787662895; cv=pass; d=google.com; s=arc-20260327; b=W45FaAZG61eOQGFyP2Hoi13WdUYKPMgRXgINV+Ju02IHfDbjSig8aKkuvw5P71uqK 2oLzUtEjg1MlFmZ5ANGTO+AEHz014trulvUjuelTk8pxQT/EayVohP61AB0VcRzG5G 5wbnHIUIqjMrUcLvrRHg67XhmyI0DYL5mv4ASurxYFVT6ZTQCFD40J3/eqKpkMIHwAsB rcdXxYpSRNG/Fr9154KQ7ff3vIyJhuteK/41Yp+2tTgcddZVE0FCTT20R/dK2HEWZj jhkO/dTO2E/y2 +dAVBdXwCzYoA1H1W30/6IKs9bClpv+iq9MDAwKqy69HY214PquKsZ hdaO==
ARC-Message-Signature: i=4; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:to:subject:message-id:date:from:in-reply-to :references:mime-version:dkim-signature; bh=xamsobc8g+24HJQFSf09vaVhNkH49/GRIZricuWAAM=; fh=Th9v4pSU48wv4z2HhpN33SBOZTP8dnv7WqCNKfUJ5K0=; b=001EK5Mn2Fm48/s3ca192S0z7kgZwdjdxYETXf6A3b1ljQp5DT1T4xh7kZmfVHKbz 1yBf1Wd5E5QAMuXjTBNeyzPob/VsZa9/uSrMoVtu7jUPKq+2P3Zj3jBlx8FQ43CgBBeV ol/zM2xqSol7l6m9H78LlrczJWd1Fy57ElECaksgCehAbojboxYfFzJzFugpHQziqFU rJsc3z0716RN52xhxvz0k6CRKyZ4ZfScf1x1315TNRq0AWykoWN/IEMTz2P1l+Im yohck6jTfAFNHyHFHrPVYwyqyFexJP37BSu7WqTGD107Ztg6RRHfiYuStj+4ub5r76 e1rw==; dara=google.com
ARC-Authentication-Results: i=4; mx.google.com; dkim-pass header.i=@klu.ac.in header.s=google header.b=Jo0p3hi8; arc-pass (i=3 spf=pass spfdomain=klu.ac.in dkim=pass dkdomain=klu.ac.in
  
```

Figure 4.6: Complete raw email headers analysis

Step 9 — Geolocation and WHOIS IP Lookup of Sending Server

The sending relay IP address 209.85.220.69 was queried through an online IP lookup service, confirming:

- **Owner / ISP:** Google LLC
- **Autonomous System:** AS15169
- **CIDR Block:** 209.85.208.0/20
- **Location:** Utica, New York, United States

The IP address 209.85.220.69 is owned by Google LLC and is located in Utica, New York, United States.

Ownership and Network Details

- **Owner/ISP:** Google LLC
- **ASN (Autonomous System Number):** AS15169
- **CIDR Block:** 209.85.208.0/20 (some sources list 209.85.192.0/18)
- **IP Type:** Public IPv4
- **Hostname:** Associated with Google infrastructure, no specific websites detected running directly on this IP [LookIP.net +3](#).

Geolocation

- **City:** Utica
- **State:** New York
- **Country:** United States
- **Time Zone:** Eastern Time Zone (Olson TZ database)
- **Accuracy:** Approximate; IP geolocation is based on ISP registration and routing information, not the exact physical location of a device [WhatIsMyIP.com +1](#).

Figure 4.9: IP lookup and geolocation analysis for sending MTA

Step 10 — Evaluating Sender Policy Framework (SPF) Status

The SPF authentication record in the header was verified: `spf=pass` (google.com: domain of `hodcse@klu.ac.in` designates 209.85.220.41 as permitted sender).

```
Content-Type: multipart/mixed; boundary="00000000000050dbb40659deb328"
-Original-Sender: hodcse@klu.ac.in
-Original-Authentication-Results: mx.google.com;
  dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s;
  arc=pass (i=1); spf=pass (google.com: domain of hodcse@klu.ac.in designates 209.85.220.41 as permitted sender);
  dmarc=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in; dara=neutral header.i=@klu.ac.in
Precedence: list
Mailing-list: list 2024cse@klu.ac.in; contact 2024cse+owners@klu.ac.in
List-ID: <2024cse.klu.ac.in>
```

Figure 4.10: Sender Policy Framework (SPF) verification in raw header

Step 11 — Verifying DomainKeys Identified Mail (DKIM) Signature

The DKIM cryptographic signature header was evaluated: DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; confirming signature validity and body hash integrity.

```
ARC-Authentication-Results: i=3; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s; arc=pass (i=1); spf=pass (google.co
hodcse@klu.ac.in designates 209.85.220.41 as permitted sender) smtp.mailfrom=hodcse@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.fro
dara=neutral header.i=@klu.ac.in
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; t=1788267693; x=1788267693; darn=klu.ac.in; h=list-unsubscribe:list-arc
help:list-post:list-id :mailing-list:precedence:x-original-authentication-results :x-original-sender:content-type:to:subject:message-id:date:from :i
to:references:mime-version:from:to:cc:subject:date :message-id:reply-to:content-type; bh=xamsobc8g+24HJQFSF09vaVhnhK49/GRIzriCuWAAm=;
b=Jo0pH3i8x2p1622HigVPh+sD0X/hDhC0EKhrQcMiZWyIVynpktGCEDUKSILrnpAR2 x+pUzIove/sRQTPop++XPPmJKGZv5PCTLbZ9ypCCU19781gESWYXA5zHbZhdLttqD4e
```

Figure 4.11: DKIM cryptographic signature header inspection

Step 12 — Examining Message-ID and ARC Authentication Seal

The ARC (Authenticated Received Chain) signature and message-id header references were analyzed to confirm end-to-end chain verification across forwarding hops.

```
ARC-Seal: i=3; a=rsa-sha256; t=1787662894; cv=pass; d=google.com; s=arc-20260327; b=CaMjLNQL+NNVEAZrKtnvgv7UwI8p+/BA5QD1GDRrQ5W8Iuv0Hs00A54QummF0icG7
jBqroiQyK+Q3NWE4AKkhm/kP4hCEGI2X1VPRIaf3nBNPSXu+FObmMqIhrJ0Dr7m/SH_vh/ohjVaguAmzHXQ9wi7ZvAyl4YKPhOSN6KMRkoaR4KPSM1ts170tEsR2tzYEdsQK8Ci
e9hG3Jx2dcREH1qXT7AAyK13zjtfJmQl0j2Dhzh1PT2c0ORh8qj48y5U7+1xdFQqE67 j/m2gt9fjLvetBE5049RI8u+fMILTskH1W5Za19capyWQ6Lw5enii49dazwCy0TBACB/x xVdQ==
ARC-Message-Signature: i=3; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-
list:precedence:to:subject: :message-id:date:from:in-reply-to :references:mime-version:dkim-signature; bh=xamsobc8g+24HJQFSF09vaVhnhK49/GRIzriCuWAAm=;
Fh=TH9vp4SI48wv4z2HhplJ3S80ZTP8dmv7MqCNKFUJ5K0-; b=ImBYcVXXevQP8cEa/3k3eded9mZrmXfUgJxsyhdYK7eRqLhXmzhbvYiBnEGQORY5
uNu/yKx8Y0SpQd3j0c4dzt1JXfbhY0ZSpRQXhMgkMGdKpNOIDocwRvfnsrmSVzklazSc+ QDAU2eq81qc804quV/EwCEKCAuPgZ1CdJQzj8oPIK2mcDKvAw3jucZVN11Iow839Pa
FuEILPM2PcBfiE+hzV/nKuILBhcMgUsU5wK5CoM/JR1nbIMI19XzrtiCm1Hui26 5jFDFBm0pFP0x6Dcd7QvPNejGuQVZ2L/1KANCQY6hf8saFdXQ8QtkjyFDIMWagYQ9H SzaQ==; darn=klu.ac.in
ARC-Authentication-Results: i=3; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s; arc=pass (i=1); spf=pass (google.com; domain of
hodcse@klu.ac.in designates 209.85.220.41 as permitted sender) smtp.mailfrom=hodcse@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in;
```

Figure 4.12: Message-ID and ARC authentication seal verification

OBSERVATIONS

1. Message traveled through authorized Google Workspace mail transfer agents in 23 seconds.
2. SPF authentication evaluated to PASS for sending IP 209.85.220.69 / 209.85.220.41.
3. DKIM cryptographic signature was signed by d=klu.ac.in and validated as PASS.
4. DMARC policy evaluated to PASS.
5. IP lookup for 209.85.220.69 verified ownership by Google LLC (AS15169).

FINDINGS

- **Authentication Alignment:** Complete alignment between the visible sender (hodcse@klu.ac.in), SPF permitted sender IP block, and DKIM cryptographic signature (d=klu.ac.in) verifies email authenticity.
- **Absence of Spoofing:** No anomalous routing hops, header modifications, or domain impersonation indicators were present.

RESULT

Successfully Completed. The email header was extracted, parsed, and verified. SPF, DKIM, and DMARC checks confirmed message authenticity without spoofing.

CONCLUSION

Header analysis confirmed that the email originated from an authorized Google Workspace mail server for domain klu.ac.in. Cryptographic authentication mechanisms (SPF, DKIM, DMARC) verified header and content integrity.

REFERENCES

- RFC 5322 (Internet Message Format)
- RFC 7208 (Sender Policy Framework)

- RFC 6376 (DomainKeys Identified Mail)
- RFC 7489 (DMARC)
- Digital Forensics Laboratory Manual: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)